

Lab Assignment 3

Timing Analysis and Constraint-Based Optimization

Problem 1: Baseline Design and Timing Analysis

Aim

Design a 4-bit Arithmetic Logic Unit (ALU) that supports:

- Addition
- Subtraction
- AND
- OR

Specifications:

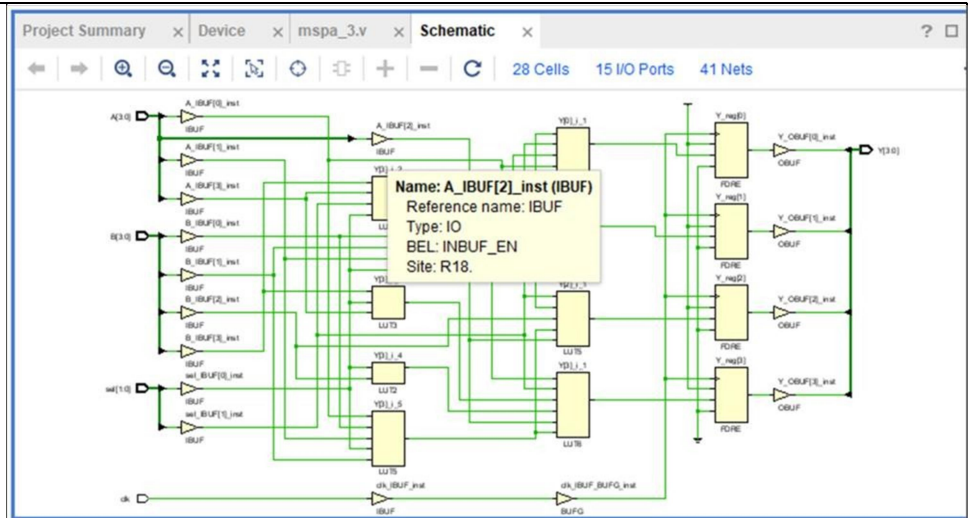
1. Write Verilog HDL code for the ALU.
2. Develop a testbench and simulate the design.
3. Perform synthesis without any timing constraints.
4. Generate a timing report.

Verilog Code:

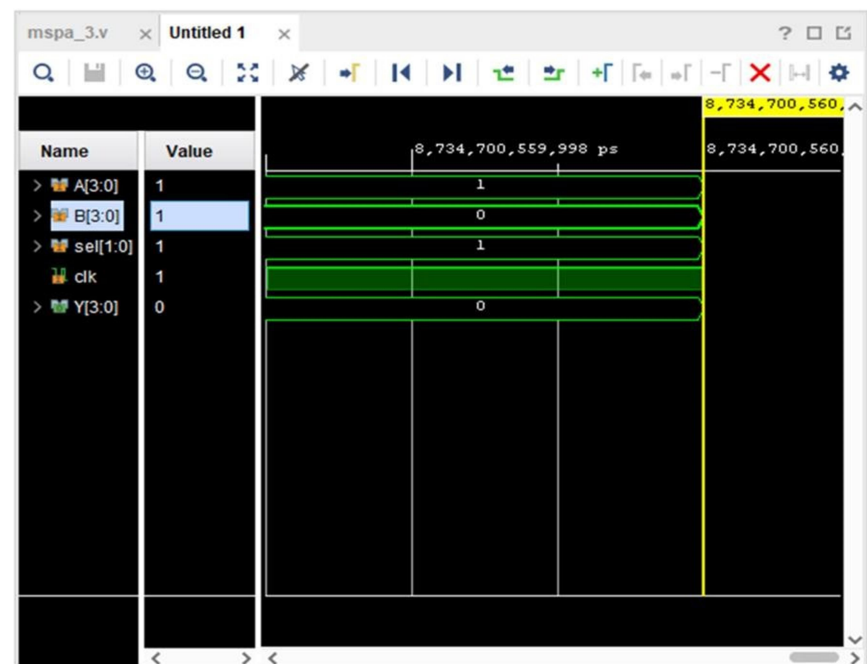
```
1 timescale 1ns / 1ps
2
3 module mspa3(
4     input [3:0] A,
5     input [3:0] B,
6     input [1:0] sel,
7     input clk,
8     output reg [3:0] Y
9 );
10
11 always @(posedge clk)
12 begin
13     case(sel)
14         2'b00: Y <= A + B;
15         2'b01: Y <= A - B;
16         2'b10: Y <= A & B;
17         2'b11: Y <= A | B;
18     endcase
19 end
20
21 endmodule
```

```
22
23 module mspa3_tb;
24     reg [3:0] A;
25     reg [3:0] B;
26     reg [1:0] sel;
27     reg clk;
28
29     wire [3:0] Y;
30
31     alu uut(
32         .A(A),
33         .B(B),
34         .sel(sel),
35         .clk(clk),
36         .Y(Y)
37     );
38
39
40     always #5 clk = ~clk;
41
42     initial
43     begin
44         clk = 0;
45
46         A = 4'b0011; B = 4'b0001; sel = 2'b00; #10; // Add
47         A = 4'b0101; B = 4'b0010; sel = 2'b01; #10; // Sub
48         A = 4'b1100; B = 4'b1010; sel = 2'b10; #10; // AND
49         A = 4'b1100; B = 4'b1010; sel = 2'b11; #10; // OR
50
51         $finish;
52     end
53
54 endmodule
```

RTL View



Wave Form



Timing Report

The screenshot shows the 'Design Timing Summary' window in Xilinx Vivado. The left sidebar contains a tree view with the following items: General Information, Timer Settings, Design Timing Summary (selected), Methodology Summary (4), Check Timing (22), Intra-Clock Paths, Inter-Clock Paths, Other Path Groups, and User Ignored Paths. The main content area displays a table with three columns: Setup, Hold, and Pulse Width. The table contains the following data:

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): inf	Worst Hold Slack (WHS): inf	Worst Pulse Width Slack (WPWS): NA
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): NA
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: NA
Total Number of Endpoints: 8	Total Number of Endpoints: 8	Total Number of Endpoints: NA

Below the table, it states: 'There are no user specified timing constraints.'

At the bottom of the window, there is a taskbar with three tabs: 'Timing Summary - impl_1 (saved)', 'Timing Summary - timing_1', and 'Timing Summary - timing_2'. A watermark 'Activate Windows Go to Settings to activate Windows.' is visible in the bottom right corner.

Result/Conclusion:

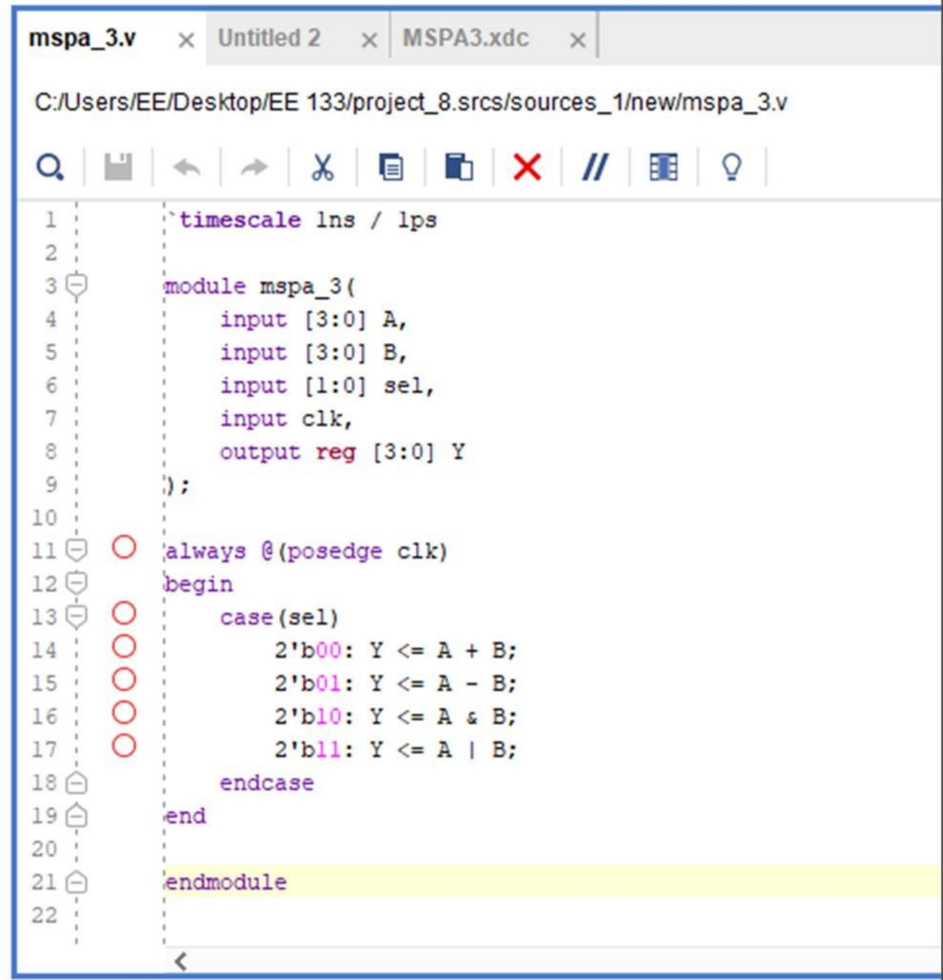
The baseline design was synthesized and implemented successfully in Xilinx Vivado. The timing report was generated to observe the initial slack, critical path delay, and resource utilization (LUTs). These results serve as the reference for comparison after applying the clock constraint.

Problem 2: Applying Clock Constraint

Aim:

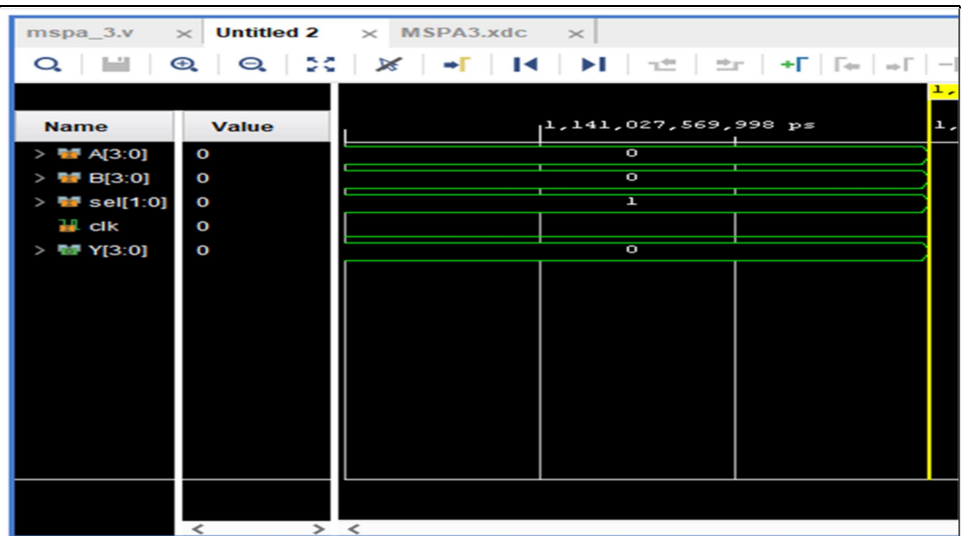
Apply a clock constraint of 10 ns (100 MHz).

Verilog Code:

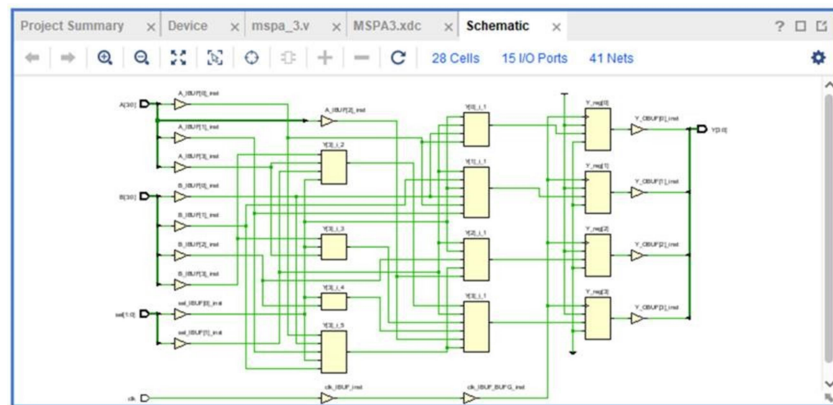


```
1      `timescale 1ns / 1ps
2
3      module mspa_3(
4          input [3:0] A,
5          input [3:0] B,
6          input [1:0] sel,
7          input clk,
8          output reg [3:0] Y
9      );
10
11      always @(posedge clk)
12      begin
13          case(sel)
14              2'b00: Y <= A + B;
15              2'b01: Y <= A - B;
16              2'b10: Y <= A & B;
17              2'b11: Y <= A | B;
18          endcase
19      end
20
21      endmodule
22
```

Waveforms:



RTL

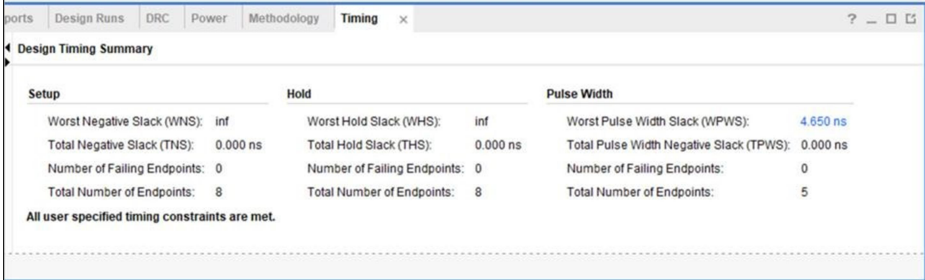


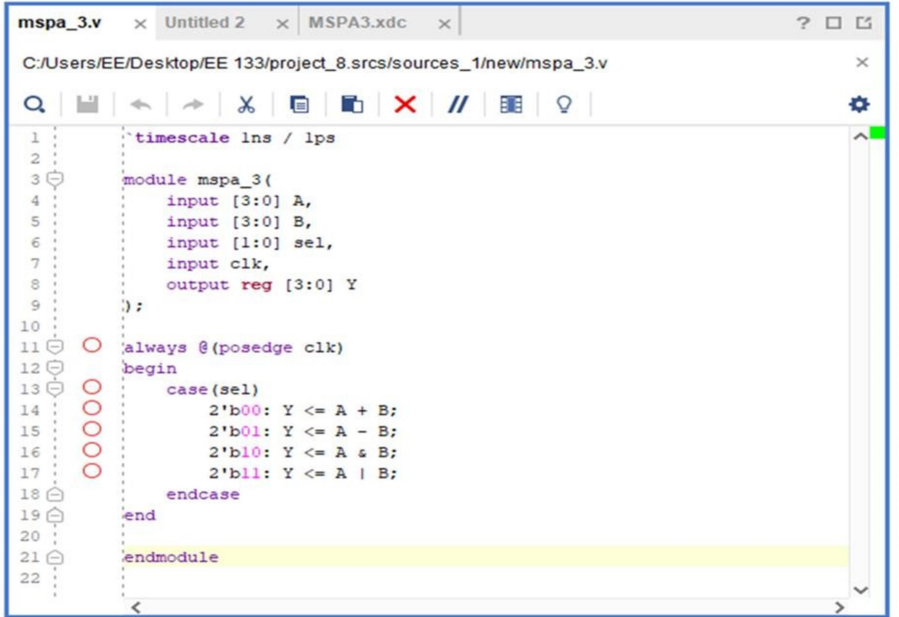
XDC file

Project Summary x Device x mspa_3.v x MSPA3.xdc x Schematic x

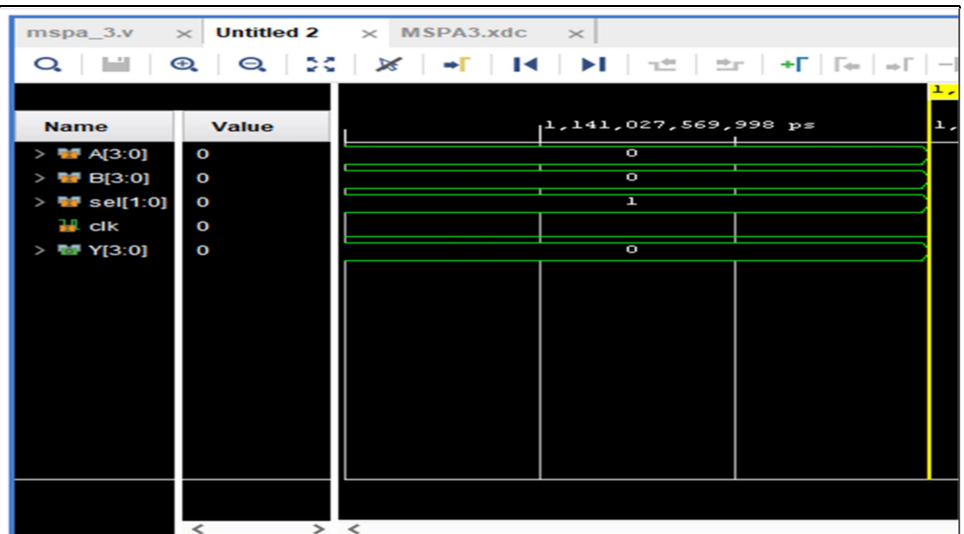
C:/Users/EE/Desktop/EE 133/project_8/srcs/constrs_2/new/MSPA3.xdc

```
1 create_clock -period 10.000 [get_ports clk]
2
```

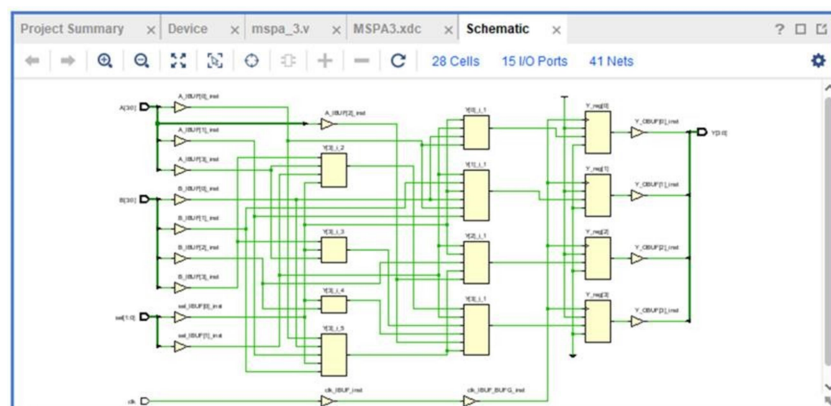
Timing Report	
Result/Conclusion:	A 10 ns (100 MHz) clock constraint was applied in the XDC file using create_clock -period 10 [get_ports clk] in Xilinx Vivado. After running synthesis and implementation again, a new timing report was generated. The slack, critical path delay, LUT usage, and maximum achievable frequency were observed and compared with the baseline design.

Problem 3: Tight Timing Constraint	
Aim:	Reduce the clock period to 5 ns (200 MHz).
Verilog Code:	 <pre> 1 timescale 1ns / 1ps 2 3 module mspa_3(4 input [3:0] A, 5 input [3:0] B, 6 input [1:0] sel, 7 input clk, 8 output reg [3:0] Y 9); 10 11 always @(posedge clk) 12 begin 13 case(sel) 14 2'b00: Y <= A + B; 15 2'b01: Y <= A - B; 16 2'b10: Y <= A & B; 17 2'b11: Y <= A B; 18 endcase 19 end 20 21 endmodule 22 </pre>

Waveforms:



RTL



XDC file

Project Summary x Device x mspa_3.v x MSPA3.xdc x MSPA33.xdc x

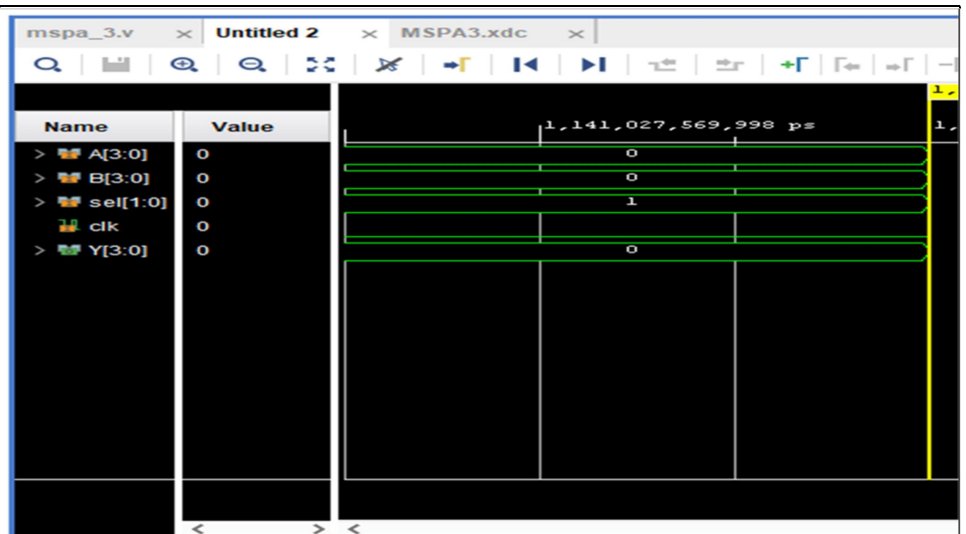
C:/Users/EE/Desktop/EE 133/project_8.srcs/constrs_3/new/MSPA33.xdc

1 create_clock -period 5 [get_ports clk]

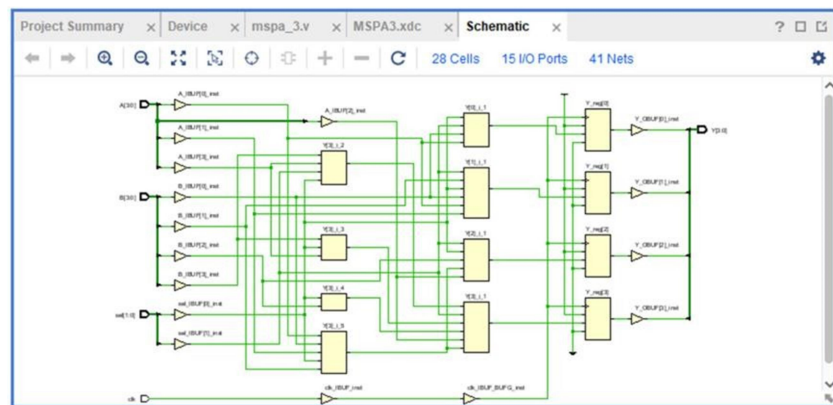
Timing Report	
Result/Conclusion:	The clock period was reduced to 5 ns (200 MHz) by updating the clock constraint in the XDC file in Xilinx Vivado. After running synthesis and implementation, a new timing report was generated. The slack decreased, the timing became tighter, and the maximum achievable frequency and critical path delay were analyzed to check if the design meets the 200 MHz requirement.

Problem 4: Input and Output Delay Constraints	
Aim:	set_input_delay 2 -clock clk [all_inputs] set_output_delay 2 -clock clk [all_outputs]
Verilog Code:	<pre> 1 timescale 1ns / 1ps 2 3 module mspa_3(4 input [3:0] A, 5 input [3:0] B, 6 input [1:0] sel, 7 input clk, 8 output reg [3:0] Y 9); 10 11 always @(posedge clk) 12 begin 13 case(sel) 14 2'b00: Y <= A + B; 15 2'b01: Y <= A - B; 16 2'b10: Y <= A & B; 17 2'b11: Y <= A B; 18 endcase 19 end 20 21 endmodule 22 </pre>

Waveforms:



RTL



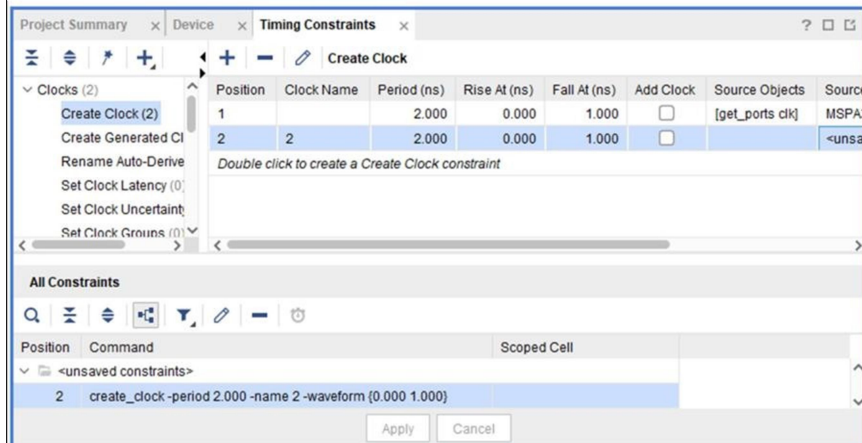
XDC file

```

Project Summary x Device x mspa_3.v x MSPA3.xdc x MSPA33.xdc x MSP4.x
C:/Users/EE/Desktop/EE 133/project_8.srcs/constrs_4/new/MSP4.xdc

1 set_input_delay 2 -clock clk [all_inputs]
2 set_output_delay 2 -clock clk [all_outputs]
  
```

Timing Report



ports

Design Runs

DRC

Methodology

Power

Timing

x

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): inf	Worst Hold Slack (WHS): inf	Worst Pulse Width Slack (WPWS): inf
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWNS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 8	Total Number of Endpoints: 8	Total Number of Endpoints: 8

All user specified timing constraints are met.

Result/Conclusion:

Input delay is the time between giving an input and when processing starts, while output delay is the time between processing completion and when the result is produced.

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Signature